SoccerGMR Performance Benchmark: Baseline vs. SI-Bind Variants (E6, GEQ, E9 / CAB)

Head-to-head evaluation across 9 temporal DETR backbones on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries). Strict single seed (2023).

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Configuration**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**
Moment-DETR	1. GMR Baseline	6.68	24.28	67.94	70.20	5.47	23.48	72.08	78.91
	2. GMR + SI-Bind (E6)	**15.21**	23.66	65.52	74.32	12.58	22.21	68.83	**81.86**
	3. GMR + SI-Bind (E6) + GEQ	14.97	**16.61**	**75.33**	74.23	12.95	**15.95**	**78.51**	81.81
	4. GMR + SI-Bind (E9 / CAB)	13.35	21.60	63.61	74.93	**12.99**	19.56	66.27	79.12
	5. GMR + SI-Bind (E9) + GEQ (Ours)	13.61	17.61	70.14	**75.53**	**12.99**	17.08	73.16	81.84
EaTR	1. GMR Baseline	16.49	21.51	65.60	76.56	15.19	20.39	67.38	81.59
	2. GMR + SI-Bind (E6)	17.12	21.80	67.74	76.23	**15.35**	20.89	68.66	81.03
	3. GMR + SI-Bind (E6) + GEQ	15.54	**17.51**	72.25	76.57	14.36	**16.42**	**76.22**	**83.05**
	4. GMR + SI-Bind (E9 / CAB)	17.17	21.58	69.33	**77.21**	15.25	19.81	72.41	82.55
	5. GMR + SI-Bind (E9) + GEQ (Ours)	**17.23**	17.80	**73.23**	75.93	15.19	16.43	75.87	82.47
QD-DETR	1. GMR Baseline	14.62	20.23	71.48	**77.01**	12.65	19.05	72.49	81.75
	2. GMR + SI-Bind (E6)	16.30	20.65	73.71	76.63	13.20	19.84	75.91	82.47
	3. GMR + SI-Bind (E6) + GEQ	**18.27**	**15.94**	**75.14**	76.16	**14.98**	15.70	**78.45**	82.73
	4. GMR + SI-Bind (E9 / CAB)	16.77	24.35	59.41	75.27	14.96	22.68	61.60	81.78
	5. GMR + SI-Bind (E9) + GEQ (Ours)	17.65	16.73	73.03	75.47	14.85	**15.65**	76.11	**82.98**
CG-DETR	1. GMR Baseline	18.65	25.66	57.47	76.40	14.67	23.28	63.23	81.81
	2. GMR + SI-Bind (E6)	17.38	22.89	70.85	75.92	15.54	21.61	73.14	80.94
	3. GMR + SI-Bind (E6) + GEQ	17.66	**16.30**	**75.64**	76.40	15.47	**15.74**	**78.88**	82.36
	4. GMR + SI-Bind (E9 / CAB)	18.76	21.55	69.27	**77.50**	**17.41**	20.21	70.53	81.24
	5. GMR + SI-Bind (E9) + GEQ (Ours)	**19.34**	16.84	75.02	77.32	16.91	15.77	77.41	**83.57**
BM-DETR	1. GMR Baseline	4.00	17.59	**75.00**	71.55	3.65	17.62	78.58	79.11
	2. GMR + SI-Bind (E6)	7.72	20.91	73.40	70.18	7.03	20.30	77.47	79.27
	3. GMR + SI-Bind (E6) + GEQ	6.17	13.19	**75.00**	70.57	5.63	13.44	**78.86**	79.08
	4. GMR + SI-Bind (E9 / CAB)	10.72	23.09	70.85	72.73	9.54	21.78	75.34	**80.12**
	5. GMR + SI-Bind (E9) + GEQ (Ours)	**11.87**	**12.07**	**75.00**	**73.43**	**10.91**	**11.71**	78.82	80.04
BAM-DETR	1. GMR Baseline	23.25	19.82	70.43	78.66	20.62	18.37	71.79	80.99
	2. GMR + SI-Bind (E6)	23.78	20.40	68.96	78.37	21.78	19.07	69.28	78.69
	3. GMR + SI-Bind (E6) + GEQ	23.99	17.48	71.31	77.08	21.56	16.37	73.81	**83.35**
	4. GMR + SI-Bind (E9 / CAB)	**24.21**	20.68	69.53	**78.96**	**22.17**	19.15	69.12	79.03
	5. GMR + SI-Bind (E9) + GEQ (Ours)	23.26	**16.16**	**71.76**	77.07	21.43	**15.03**	**74.57**	82.86
Sim-DETR	1. GMR Baseline	20.73	22.51	74.69	72.97	19.18	21.78	**75.72**	78.87
	2. GMR + SI-Bind (E6)	21.17	21.26	74.89	75.12	20.08	20.83	73.10	77.83
	3. GMR + SI-Bind (E6) + GEQ	21.14	12.72	**75.00**	**75.60**	19.96	12.59	75.25	81.21
	4. GMR + SI-Bind (E9 / CAB)	21.34	23.16	72.12	74.70	20.46	22.30	73.46	79.47
	5. GMR + SI-Bind (E9) + GEQ (Ours)	**22.45**	**11.71**	**75.00**	74.86	**21.05**	**11.60**	75.52	**81.43**
UVCOM	1. GMR Baseline	7.12	17.24	75.00	70.03	6.40	16.83	**76.80**	78.98
	2. GMR + SI-Bind (E6)	18.99	16.30	**75.40**	76.91	17.74	15.55	73.11	75.20
	3. GMR + SI-Bind (E6) + GEQ	17.94	17.89	72.92	77.37	16.55	17.04	73.98	**81.37**
	4. GMR + SI-Bind (E9 / CAB)	**19.74**	17.28	72.89	**77.84**	**18.58**	16.69	69.85	75.33
	5. GMR + SI-Bind (E9) + GEQ (Ours)	18.51	**14.78**	71.55	77.77	17.37	**14.36**	69.25	77.51
FlashVTG	1. GMR Baseline	**25.43**	**25.47**	**66.35**	**74.30**	**20.43**	**24.13**	**68.48**	**82.21**
	2. GMR + SI-Bind (E6)		*-- running / pending --*				*-- running / pending --*		
	3. GMR + SI-Bind (E6) + GEQ		*-- running / pending --*				*-- running / pending --*		
	4. GMR + SI-Bind (E9 / CAB)		*-- running / pending --*				*-- running / pending --*		
	5. GMR + SI-Bind (E9) + GEQ (Ours)		*-- running / pending --*				*-- running / pending --*		

Method 1 (GMR Baseline): Official temporal DETR baseline checkpoints evaluated with native bipartite matching.
Method 2 (GMR + SI-Bind [E6]): Structural Triad (Temporal Residual Stream + Slot-Private Innovation + Dominance-Aware Refresh).
Method 3 (GMR + SI-Bind [E6] + GEQ): Two-stage Global Event Query fine-tuning on top of SI-Bind E6 representations.
Method 4 (GMR + SI-Bind [E9 / CAB]): Serial C -> A -> B Triad (Controlled Refresh -> Temporal Re-binding -> Context Innovation).
Method 5 (GMR + SI-Bind [E9] + GEQ [Ours]): Two-stage Global Event Query fine-tuning on top of SI-Bind E9 representations (Val Orc-gap selected).
Orc-Gap (%): Area between Oracle envelope and G-mIoU@1 threshold curve across tau in [0, 1]. Lower is better.
Bold values highlight the optimal performance across methods per backbone. FlashVTG E6/E9 currently in queue.